10) Sensorless Corridor Robot:

```
        belief = {1, 2, 3, 4}

def move_right(belief):
    return {min(cell + 1, 4) for cell in belief}

def move_left(belief):
    return {max(cell - 1, 1) for cell in belief}

print("Initial belief:", belief)

for step in range(3):
    belief = move_right(belief)
    print("After Right", step + 1, ":", belief)

if belief == {4}:
    print("Robot is certainly at cell 4!")
else:
    print("Robot location is still uncertain.")

print("\nWith a bump sensor:")
position = None

for cell in range(1, 5):
    # Move left until a wall bump identifies cell 1.
    if cell == 1:
        position = 1
        break

print("After identifying the left wall, move Right 3 times.")
print("Final position: cell 4")
```